

Applied Human Computer Interaction (AHCI)

Asif Fazwani

Chapter 9 – Evaluation Techniques

What is Evaluation?

Evaluation = **Assessing how well a system works**

Does it meet: User needs? Design expectations?

“Did we build the right thing, the right way?”

Why Evaluation is Important It ensures: Usability, Effectiveness, User satisfaction. If Instagram releases a new feature:

- Do users understand it?
- Do they actually use it?

Evaluation is NOT a Final Step It happens throughout design, continuous process. Test → Improve → Test again

Evaluation in Design Cycle It happens during design, during prototyping, after implementation Feedback loop: Evaluation → Redesign → Improve

Types of Evaluation

- **Expert-Based Evaluation**
- **User-Based Evaluation**

Expert-Based Evaluation It is carried by: Designers / usability experts and No real users required. It is best for early designs and quick feedback

User-Based Evaluation It is carried by: Real users, they observe actual usage

Example: Watching users navigate TikTok

Goals of Evaluation

Evaluation has **3 main goals**:

- Assess functionality
- Assess user experience
- Identify problems

“Does it work, feel good, and avoid issues?”

Goal 1 – Assess Functionality

- Does the system: Support user tasks? Provide required features? Is functionality: Easy to access?

Example Searching in Amazon → Users expect: Search by product name & Filter options

If missing → system fails functionally

System should match: How users think

Example: Maps app should allow search by: Location name, Address (Like in Google Maps)

Measuring Performance, It evaluates **Speed, Accuracy, Task completion** how fast users find a product

Goal 2 – Assess User Experience

How does it feel to use? It includes **ease of learning** , **usability**, **satisfaction**

Example Scrolling in TikTok

Smooth, Addictive, Easy to use (Good UX = users stay longer)

Cognitive Load Avoid too much thinking and too much memory load

Emotional Response Users may feel: Enjoyment, Frustration, Satisfaction (Fun filters in Snapchat)

Goal 3 – Identify Problems

Find: Confusing parts, Errors, Unexpected behavior

Example Checkout failure in Amazon

- User clicks “Buy Now”
- Nothing happens

Critical usability issue

Focus on Trouble Spots Identify: where users struggle

- Fix: Specific issues, improves overall system

Evaluation Through Expert Analysis

Types of Expert Analysis

- Cognitive Walkthrough
- Heuristic Evaluation
- Models
- Previous Work

Cognitive Walkthrough

Cognitive Walkthrough Step-by-step evaluation of: User tasks

It focuses on **ease of learning**. “Can a new user figure this out?”

It’s **main focus** is learning by exploration not training or manuals

What Evaluators Do Simulate: User actions step-by-step

Ask: Will user succeed at each step?

Expert ‘walks through’ design to identify potential problems using psychological principles

For each task walkthrough considers

1. What impact will interaction have on user?
2. What cognitive processes are required?
3. What learning problems may occur?

Analysis focuses on goals and knowledge: does the design lead the user to generate the correct goals?

Cognitive Walkthrough

Example: Post a story on Instagram

Steps: Open app → Click “+” → Select story → Upload

Key Questions

1. **Goal Match** Does action match user goal? “Is this what I want to do?”
2. **Visibility** Can user see the option? “Can I find it?”
3. **Recognition** Will user understand it? “Is this the right button?”
4. **Feedback** Does system respond clearly? “Did it work?”

Example (Story Upload)

- Button hidden?
- Icon confusing?
- No confirmation?

All = usability issues

Heuristic Evaluation

Proposed by Nielsen and Molich.

Uses: Rules of thumb (heuristics) Purpose: Find usability problems

Evaluating login screen: No error message when password is wrong

Why Use It? Works for: Early designs, Prototypes, Final systems

How It Works Multiple experts: Evaluate system they work independently and then combine results later.

What Do Evaluators Do? Check system against 10 usability (Nielsen) heuristics and identify violations (problems)

Heuristic Evaluation - Severity Rating (0–4)

Score	Meaning
0	I don't agree that this is a usability problem at all
1	Cosmetic problem only: need not be fixed unless extra time is available on project
2	Minor usability problem: fixing this should be given low priority
3	Major usability problem: important to fix, so should be given high priority
4	Usability catastrophe: imperative to fix this before product can be released (Nielsen)

Severity Factors

- How frequent?
- How serious?
- Easy to fix?
- Persistent issue?

After Evaluation

Combine: All findings

Prioritize: High severity issues

Nielsen's 10 Heuristics

- 1. Visibility of system status:** Always show: What's happening.
Upload progress in Instagram
- 2. Match between system and the real world:** Use familiar language "Add to Cart" in Amazon.
- 3. User control & freedom:** Provide: Undo / Cancel. "Undo send" in email apps
- 4. Consistency & Standards:** Same actions = same results. Like button works same everywhere
- 5. Error Prevention:** Prevent mistakes. Disable "Submit" if form incomplete

Nielsen's 10 Heuristics

- 6. Recognition rather than recall** Show options instead of making users remember suggested searches in Google
- 7. Flexibility and efficiency of use:** Support beginners + experts
Keyboard shortcuts
- 8. Aesthetic and minimalist design:** Keep it clean and simple.
Simple UI of WhatsApp
- 9. Help users recognize, diagnose and recover from errors** Clear error messages “Incorrect password” + reset option
- 10. Help and documentation** Easy help → Help center in apps

Model-Based Evaluation

Uses: **Cognitive & design models**

Purpose: Predict user behavior and Evaluate design before building

“Use theory to predict usability instead of testing user

Designing a search feature in YouTube

Model predicts: Users will: Tap search icon, Type keywords, Select suggestion

- Count taps + typing time
- Estimate total time to find a video

Model-Based Evaluation

Types of Models Used

1. GOMS model
2. Keystroke-Level Model (KLM)
3. Design rationale
4. Dialog models

GOMS Model Goals Operators Methods Selection. It predicts how users perform tasks

GOMS compares methods based on: Number of steps (operators), Time taken, Complexity. It **breaks each method into small actions** and compares them

Example Searching product on Amazon

Goal: Find product Method 1: Search bar Method 2: Browse categories

Model helps choose: Faster / better method

Model-Based Evaluation

Keystroke-Level Model (KLM) Predicts: Time taken for actions and measures: Clicks, Typing, Delays

KLM Example: Sending a message on WhatsApp
Open chat → Type message → Press send → Model estimates: Total time taken

Design Rationale Evaluates design decisions based on Evidence + reasoning. It helps choose the best design option

Example: Button placement in Instagram

Option A: Top-right Option B: Bottom-center

Choose based on: Ease of reach & User behavior

Dialog Models Evaluate interaction flow and detect **Dead ends, Loops, Complexity**

Example: Checkout flow in Daraz Add to cart → Payment → Confirmation

Problem: User stuck in payment loop

Activity 8

<https://tinyurl.com/Spring2026-Activity8>

Activity 8

Problem 1: New User Confusion

Method: Cognitive Walkthrough

Why: focuses on Learnability and first-time user experience. It evaluates Step-by-step actions.

Problem 2: UI Issues

Method: Heuristic Evaluation

Why: It identifies usability problems and violations of design principles, fast and cost-effective

Performance Issues

Method: Model-Based Evaluation

Why: It predicts time taken and efficiency. Also compares different design options.

Activity 8

In what ORDER would you apply these methods?

1. Heuristic Evaluation → fix obvious UI issues
2. Cognitive Walkthrough → improve learnability
3. Model-Based Evaluation → optimize performance

Can one method solve all problems?”

Answer: No, each method focuses on different aspects:

Cognitive → learning

Heuristic → usability

Model → performance

Evaluation Through User Participation

Testing system with **real users (not just experts)**. It happens usually after **prototype is ready**

Why Important? Experts $\neq$ real users. It reveals **actual behavior, confusion, emotions**.

Example: Testing a new **food delivery app**. Real users: Get confused at checkout, they prefer “Cash on Delivery” option. Designers wouldn’t catch this without users.

When Do We Use It? Typically used after **prototype or working system, but also useful for:** early stage $\rightarrow$ requirements gathering

Example: Watching how students use LMS (Learning Portal). University tests new LMS: Students complain: “Too many clicks to upload assignment”

Evaluation Through User Participation

Two Styles of Evaluation

1. Laboratory Studies

Laboratory Studies (Controlled Testing)

- Controlled environment (lab / quiet room)
- No distractions and advanced tools (screen recording, eye tracking)

Pros: Accurate measurements and easy comparison of designs

Cons: Unrealistic environment

Example Testing a **gaming UI**

- In lab → player fully focused
- In reality → distracted by messages, calls

2. Field Studies

Field Studies (Real Environment)

- Conducted in **user's real environment**
- Observe real-life usage

Pros Real behavior, Context matters (interruptions, multitasking)

Cons Noisy, messy, hard to control

Example Testing a **study app** at home: WhatsApp notifications
Parents interrupting
Reveals real usability issues

Experimental Evaluation

A **controlled experiment** to test usability to provides **real data (evidence-based)**. It's main purpose to test a **hypothesis** and compare different **design options**

- Evaluator chooses hypothesis to be tested
- Number of experimental conditions are considered which differ only in the value of some controlled variable.
- Changes in behavioral measure are attributed to different conditions

Experimental Factors: 1. Participants 2. Variables 3. Hypothesis 4. Experimental Design

Experimental Factors

Participants: Must match **real users** and they have similar age bracket, experience and Tech knowledge

Example: Testing a **food app**. Don't test only developers. Test real users (students, workers, etc.)

Sample Size:

- Small users → find problems
- Large users → reliable results

Rule of Thumb

- 5 users → find usability issues
- 10+ users → better for experiments

Example Testing new Instagram feature:
3 users ≠ reliable
20 users = better insights

Experimental Factors

Variables:

Independent Variable (IV): What you **change**. Interface style, number of menu items

Dependent Variable (DV): What you **measure**. Time taken, number of errors.

Example: IV: Button color (Red vs Green) DV: Click rate / time to click

IV: Menu size (5 vs 10 items) DV: Time to find option

Experimental Factors

Hypothesis: A **prediction** about results

“Users complete tasks faster with fewer menu items”

Prediction of outcome, framed in terms of IV and DV

Example: “error rate will increase as font size decreases”

Null Hypothesis: “No difference between designs”

Goal: **Prove null hypothesis wrong**

Example: null hypothesis = “no change with font size”

Experimental Factors

Experimental Design: Between-Subjects Design & Within-Subjects Design

Between-Subjects Design

How it works Different users → different conditions

Example Group A → Dark Mode Group B → Light Mode

Pros: No learning effect.

Cons: Needs more users, user differences can affect results

Within-Subjects Design:

How it works Same users try **all conditions**

Example: Same user uses: Dark mode and Light mode

Pros: Fewer users needed, More consistent

Cons: Learning effect

Statistical measures

It helps **interpret experiment results** and turns data → **meaningful insights**

Before you start to do any statistics: Look at data and save original data

Types of Data Variables:

Discrete Variables (Limited values) Example: Button Color (Red/Green), Menu type, Theme (Dark/Light)

Continuous Variables Any value possible

Examples: Time taken, User rating, Time to upload reel

Statistical measures

Statistical measures - Data Distribution

Normal Distribution (Bell Curve) Most common pattern, Symmetrical shape

Parametric Tests it is used when data is normal (bell-shaped)

Advantages: More **accurate & powerful**

Example: Measuring: Time to send message, usually follows normal distribution

Non-Parametric Tests: It is used when data is not normal. It uses **ranking instead of exact values**

Example Scores: 57, 32, 61, 49 Convert to ranks: 3rd, 1st, 4th, 2nd

Pros: Safer, handles outliers **Cons:** Less powerful

Contingency Tables: Classify data by discrete attributes, Count number of data items in each group.

Device	Success	Fail
Mobile	20	5
Laptop	15	10

Experimental studies on groups

Evaluating group systems $\neq$ single-user systems. It's much more complex.

Why? Human + Human interaction, Not just Human + Computer

Example: Testing a group project tool (like Google Docs / Discord)
→ Behavior depends on teamwork, not just UI

Major Challenges

1. **Participant Group:** More participants needed
2. **Experimental Task:** choice of task, choosing a suitable task.
3. **Data Gathering:** Longer experiments
4. **Analysis:** Complex data

Example:

10 individual tests = 10 people

Group of 3 → need 30 people

Experimental studies on groups

Participant Challenges

Problems:

- Need many users
- Coordination is difficult
- Group dynamics vary

Example: Group assignment:

- Group 1 → all active
- Group 2 → 1 leader, others silent
- Group 3 → chaos

Same system, different outcomes

Experimental studies on groups

Experimental Task/Task Design:

Tasks must: Require collaboration, Not be doable alone, Use the system fully

Types of Tasks

- 1. Creative Tasks:** Write report / proposal Example: Students create startup idea using shared doc
- 2. Design Tasks:** Redesign something Example: Redesign a university website layout using collaborative tool
- 3. Decision Games** Require agreement Example: “Desert survival game” → Rank items together
- 4. Conflict + Cooperation Tasks:** Mixed interests Example: Multiplayer strategy game (like team-based battle games)
- 5. Time-Critical Tasks:** Fast decision-making. Example: Team managing a simulated delivery system (like food orders rush hour)

Experimental studies on groups

Data Gathering:

Problem: Multiple users = multiple recordings

It needs video per user, audio, screen logs. Example: 3 users → 6+ video record

Solution for Data Collection Approach: Record each user individually

Capture:

- What they see
- What they hear
- What they do

Example: Like recording each player's POV in a multiplayer game.

Experimental studies on groups

Analysis Challenges: Groups behave differently. Results become unpredictable.

Example: One group collaborates well → high score.

Another group argues → low score (Not because of system, but people!)

Statistical Difficulty

- Hard to get consistent results
- Need many groups
- Often small sample size

Results may NOT be statistically significant

Reality: Even good systems may look “bad” due to group behavior.

Solutions to Analysis Problems

- 1. Within-Group Design:** Same group tests multiple conditions
- 2. Micro-analysis:** Study small details. Example: Time gaps in conversation (Who speaks more)
- 3. Anecdotal Analysis:** Focus on key events. Example: “User got confused here”
“Team argument happened here”.

Field Studies with Groups

Experiments dominated by group formation

Field studies more realistic: distributed cognition $\Rightarrow$ work studied in context real action is situated action

Physical and social environment both crucial

Problem with lab: Artificial environment

Better: Real-world observation

Example: Observe students collaborating in: Classroom, Hostel, Zoom meeting

Observational Techniques

Watching users interact with a system. Recording what they do

“Don’t ask users what they do, watch what they actually do”

Example: Watching how students use a food delivery app

→ Do they search? Scroll? Get confused?

Problem with Observation Alone

You see actions, but NOT thoughts

Missing: Why user clicked something? What were they thinking ?

Example: User clicks “Back” → You don’t know WHY

Observational Techniques: 1. Think Aloud 2. Cooperative evaluation 3. Protocol analysis 4. Automated analysis 5. Post-task walkthroughs

Observational Techniques

Think Aloud Technique: User speaks while using system

It explains **what they are doing, why they are doing it.**

Example: I'm clicking here because I think this opens my cart...

Advantages:

- simplicity - requires little expertise
- can provide useful insight
- can show how system is actually use

Disadvantages:

- Users may forget to speak
- Talking changes behavior
- Subjective data

Example: User becomes slower because they are explaining everything

Observational Techniques

Cooperative Evaluation: Upgrade of Think Aloud

- User + evaluator collaborate
- User can ask questions
- Evaluator can ask “why?”

Example: Evaluator: “Why did you click that?” → User: “Because I thought it was the login button”

Benefits of Cooperative Evaluation

- More natural interaction
- Better insights
- Immediate clarification

Example: Like debugging together in a group project

Observational Techniques

Protocol Analysis: Record of user session. Includes: Actions, Speech and Behavior

Recording Methods

1. **Paper & Pencil:** Cheap but limited - Writing “User confused at step 3”
2. **Audio Recording** Captures think aloud. Problem: Can’t see actions
3. **Video Recording** Captures behavior + screen - Record face + screen interaction
4. **Computer Logging** Automatically records actions - Clicks, typing, navigation paths
5. **User Diaries** Users write their experience - “App crashed while ordering food”

Best Practice: Use multiple methods together. Video + audio + logs + notes. More data = better understanding

Big Challenge – Data Overload: Too much data, hard to analyze, Synchronization issues (Matching video + audio + clicks)

Observational Techniques

Automated Tools Help analyze data faster

Example Tools:

- EVA → Video annotation
- DRUM → Performance measurement
- Multimedia tools → Sync video + logs

Example: Tagging “user error” moments in video

Tool Limitations

- Analyst may miss events
- Hard to focus on everything
- Still requires human judgment

Observational Techniques

Post-Task Walkthrough Replay session to user. Ask them to explain actions

Example: “Why did you undo this step?”

Why Walkthrough is Important

- Reveals hidden thinking
- Explains unclear actions
- Shows alternative choices

Example: I almost clicked checkout but wasn't sure about price

When Walkthrough is Necessary

Complex tasks, Silent users, High-focus activities

Example: Gaming, coding, or exam systems

Question Time

Testing a New Shopping App

You are evaluating a **new shopping app** (like Daraz or Amazon).
Users are asked to:

- Search for a product
- Add to cart
- Checkout

Which observational technique method will you use?

Query Techniques

Ask users directly about the system get **opinions, feelings, and feedback**. “Don’t guess user experience, just ask them.”

Advantages: Direct **user perspective**, Easy & low cost, Reveals **hidden issues** designers miss.

Limitations: Subjective (opinions, not facts), Users may **forget or exaggerate**, Hard to compare designs they haven’t used.

Types of Query Techniques

1. **Interviews** One-on-one discussion. Flexible, deep insights
2. **Questionnaires** Structured questions. Reach many users quickly

Query Techniques

Interviews Scenario: Food Delivery App (like Foodpanda)

Example Questions:

- “What was the most confusing part?”
- “Why did you cancel your order?”
- “What would you improve?”

“Delivery charges show too late. Reveals **real frustration**”

Questionnaires: Example Questions: Scalar (Rating):

“Checkout process is easy”

Multiple Choice: How do you search products?, Search bar, Categories, Recommendations

Open-ended: “What should we improve?”

App: Daraz after shopping, show survey:

- Rate your experience (1–5)
- “Did you face any issue?”
- “What feature do you want next?”

Interviews Advantages:

- It can be varied to suit context
- Issues can be explored more fully
- It can elicit user views and identify unanticipated problems

Disadvantages

- Very subjective & Time consuming

Questionnaire Advantages:

- Quick and reaches large user group and it can be analyzed more rigorously

Disadvantages

- less flexible and less probing

Physiological Evaluation Methods

Measure **body signals** instead of asking users. It captures **real reactions (not just opinions)**

“Don’t ask how users feel, measure it directly.”

Advantages:

- Objective (real data, not biased)
- Detect **hidden emotions**
- Shows **cognitive load + stress**

Limitations:

- Expensive 
- Hard to interpret
- Needs special equipment

Physiological Evaluation Techniques: 1. Eye Tracking 2. Physiological Measurements

Physiological Evaluation Methods

Eye Tracking: What it measures

Where users look. How long they look. How they scan the screen

Key Metrics:

- **Fixations** → where user focuses
- **Fixation duration** → confusion level
- **Scan path** → navigation pattern

Example: Testing shopping app (like Daraz)

Observation: User keeps looking at banner ads instead of “Buy Now”. Important button is **not visually prominent**

Eye Tracking Insight

Good Design: User looks → clicks quickly

Bad Design: Eyes jumping everywhere. Long pauses meaning: more eye movement = more confusion

Physiological Evaluation Methods

Physiological Measurements - What is measured:

Heart rate → stress/anger

Sweat (GSR) → effort/stress

Muscle activity (EMG) → engagement

Brain activity (EEG) → attention

Example Gaming App (like PUBG Mobile)

During gameplay: Heart rate increases, Sweat increases. Game is **intense/exciting OR stressful**

Challenges/Problems:

Hard to map signals → exact emotions

Same signal = different meanings

Expensive labs required

When to Use? Best for advanced usability labs, HCI research in big companies, studying **attention & stress**

Choosing an Evaluation Method

So many evaluation methods which one to use? No single best method.
Choose based on **context + goals**

When in Process? Design vs Implementation

Design Stage: Quick, cheap methods, Expert-based (e.g., walkthrough)

Implementation Stage: Real testing with users. More detailed evaluation

Example: Prototype of Daraz → expert review (Live app → user testing)

Style of Evaluation: Laboratory vs Field

Laboratory: Controlled environment. Good for experiments

Field: Real-world usage. Natural behavior

Example: Lab → test checkout speed. Field → observe real shopping behavior

Choosing an Evaluation Method

Subjective vs Objective: Best Practice: Use both

Subjective: Based on opinions e.g., Think Aloud, Interviews

Objective: Measurable data e.g., Time, errors, eye tracking

Type of Data. Quantitative vs Qualitative:

Quantitative: Numbers (time, clicks, errors), Easy to analyze

Qualitative: Opinions, feelings, harder but deeper insight

“Checkout took 30 sec” → Quantitative

“Checkout is confusing” → Qualitative

Choosing an Evaluation Method

Information & Interference

Level of Information:

- Low-level → specific (button size)
- High-level → overall UX

Intrusiveness: Obtrusive → affects behavior (Think aloud),
Unobtrusive → natural (Logs)

Resources Availability: Consider Resources: Time, Budget,
Users, Equipment, Expertise

Example: Startup testing app (like TikTok clone)

Limited budget → Use heuristic evaluation + questionnaire

Big company → Use lab + eye tracking + experiments